LL97 Data Cleaning & Reduction Report

Initial Raw Data Rows: 29,842

Step 1: Standardizing Null Values

Rows Before: 29,842 | Rows After: 29,842 | Dropped: 0

Replaced the text 'Not Available' with standard system NULL (NaN) values to allow numeric filtering. This step doesn't drop rows, but prepares the data.

Step 2: Filtering Test Properties

Rows Before: 29,842 | Rows After: 29,804 | Dropped: 38

Removed properties marked as 'Test' in the Construction Status column, keeping only real assets.

Step 3: LL97 Size Threshold Filter

Rows Before: 29,804 | Rows After: 17,840 | Dropped: 11,964

Filtered properties to only include buildings with a Gross Floor Area (GFA) >= 50,000 sq ft, matching the Local Law 97 regulatory baseline.

Step 4: New York City Boundaries Filter

Rows Before: 17,840 | Rows After: 17,818 | Dropped: 22

Processed typo-ridden city names and filtered the dataset to strictly contain addresses within New York City boroughs (dropping out-of-state/invalid locations).

Step 5: Data Integrity & Alerts Filter

Rows Before: 17,818 | Rows After: 11,732 | Dropped: 6,086

Dropped records containing critical missing or overlapping meter data (e.g., energy gaps, less than 12 months of data) to ensure emissions baseline accuracy.

Step 6: Penalty Outliers Filter

Rows Before: 11,732 | Rows After: 11,724 | Dropped: 8

Calculated projected carbon penalties and removed extreme statistical outliers (Penalty > $1700/sqft) likely resulting from data entry errors.

Step 7: Zero Energy Use Filter

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Rows Before: 11,724 | Rows After: 11,719 | Dropped: 5

Dropped records where reported site energy use was exactly zero (which is physically impossible for operating buildings).

Step 8: Remove Duplicate Property IDs

Rows Before: 11,719 | Rows After: 11,640 | Dropped: 79

Removed duplicate rows based on Property ID, retaining only unique buildings in the final dataset.

Final Clean Data Rows: 11,640

Total Rows Dropped: 18,202 (61.0%)